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GitHub link: <https://github.com/alissacanesim/jupyter-exploration>

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This lab provided an opportunity to review and practice the basics of GitHub and Jupyter Notebooks. Although this was not my first time using either tool, it was helpful to revisit the fundamentals and reflect on how these technologies are used in academic and professional environments. Over the course of my Artificial Intelligence program, I have encountered both GitHub and Jupyter Notebooks in different classes and projects. However, I remember that adapting to GitHub was initially more challenging than learning how to use Jupyter Notebooks. While Jupyter felt relatively intuitive, GitHub introduced new concepts such as repositories, commits, branches, and version control, which required some time and practice to fully understand.

The first part of the lab focused on GitHub. I created a public repository called “jupyter-exploration” and initialized it with a README file. After creating the repository, I edited the README to include a short description of the lab and committed the changes to the main branch. I also spent some time reviewing the GitHub interface and exploring sections such as repositories, issues, and pull requests. Even though I have seen these features before, I appreciated having the chance to revisit them because GitHub contains many tools that become easier to understand through repeated use rather than through a single introduction.

One aspect I found particularly interesting is how GitHub serves as both a storage platform and a collaboration tool. Before starting my studies in Artificial Intelligence, I viewed it primarily as a website where programmers uploaded code. As I gained more experience, I realized that GitHub plays a much larger role in project organization, version tracking, and teamwork. The ability to keep a history of changes through commits makes it possible to document progress and return to previous versions when necessary. This feature is especially valuable in larger projects where multiple files and contributors are involved.

For the Jupyter Notebook portion of the lab, I chose to use Google Colab instead of installing Jupyter locally. Google Colab provides a browser-based environment that includes Jupyter Notebook functionality without requiring additional software installation. After creating a new notebook, I added a Markdown cell containing a title and a brief description. I then created a Python code cell and executed a simple “Hello, World!” program. Once the notebook was completed, I saved it, downloaded the .ipynb file, and uploaded it to my GitHub repository.

Although the notebook itself was simple, it highlighted one of the reasons Jupyter has become so popular in data science and artificial intelligence. Unlike traditional programming environments, Jupyter allows code, explanations, results, and visualizations to exist within the same document. This makes it easier to experiment with ideas, document thought processes, and share work with others. In my own experience, notebooks are particularly useful when learning new concepts because they encourage a more interactive and exploratory approach to programming.

Another useful aspect of this lab was seeing how GitHub and Jupyter can be used together. Jupyter Notebooks make it easy to create and test code, while GitHub provides a platform for storing, organizing, and sharing those notebooks. In many AI and data science projects, these tools are used side by side. Understanding how to connect them is therefore an important skill for students entering technical fields.

Overall, this lab served as a good review of tools that are becoming increasingly important in technology and artificial intelligence. While the activities themselves were introductory, they reinforced concepts that I will continue to use throughout my academic and professional development. Revisiting GitHub reminded me how much more comfortable I have become with version control compared to when I first encountered it, while working with Google Colab and Jupyter Notebook reinforced the value of interactive computing environments. As I continue my studies in Artificial Intelligence, I expect both GitHub and Jupyter Notebooks to remain essential parts of my workflow.